

WEEK 7 - TDPT

We saw the interaction representation last time:

$$i\hbar \partial_t |\Psi_I(t)\rangle = \hat{H}'_I(t) |\Psi_I(t)\rangle$$

Where $|\Psi_I(t)\rangle = e^{i\hat{H}_0 t/\hbar} |\Psi(t)\rangle$ and $\hat{H}'_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{H}'(t) e^{-i\hat{H}_0 t/\hbar}$

If we take the integral of this TDSE over some time t , from t_0 to t , we get:

$$\int_{t_0}^t dt, i\hbar \partial_t |\Psi_I(t)\rangle = \int_{t_0}^t dt, \hat{H}'_I(t) |\Psi_I(t)\rangle$$

$$i\hbar (|\Psi_I(t)\rangle - |\Psi_I(t_0)\rangle) = \int_{t_0}^t dt, \hat{H}'_I(t) |\Psi_I(t)\rangle$$

$$|\Psi_I(t)\rangle = |\Psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^t dt, \hat{H}'_I(t) |\Psi_I(t)\rangle$$

Though we can't exactly call this a solution, we can substitute in

$$|\Psi_I(t_1)\rangle = |\Psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^{t_1} dt_2 \hat{H}'_I(t_2) |\Psi_I(t_2)\rangle$$

on the RHS to get:

$$|\Psi_I(t)\rangle = |\Psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^t dt, \hat{H}'_I(t) \left(|\Psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^{t_1} dt_2 \hat{H}'_I(t_2) |\Psi_I(t_2)\rangle \right)$$

$$= |\Psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^t dt, \hat{H}'_I(t) |\Psi_I(t_0)\rangle + \frac{1}{(i\hbar)^2} \int_{t_0}^t dt, \int_{t_0}^{t_1} dt_2 \hat{H}'_I(t_1) \hat{H}'_I(t_2) |\Psi_I(t_2)\rangle$$

We can do this substitution infinitely, which yields:

$$|\Psi_I(t)\rangle = \sum_{n=0}^{\infty} |\Psi_I^{(n)}(t)\rangle \text{ where}$$

$$|\Psi_I^{(n)}(t)\rangle = \left[\left(\frac{1}{i\hbar} \right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \dots \int_{t_0}^{t_{n-1}} dt_n \hat{H}'_I(t_1) \hat{H}'_I(t_2) \dots \hat{H}'_I(t_n) \right]$$

all applied to $|\Psi_I(t_0)\rangle$, which is the only thing we can compute.

This is known as the Dyson Series. Note the similarity here to perturbative expansion where, e.g. the first order correction is

$$|\Psi_I^{(1)}(t)\rangle = \frac{1}{i\hbar} \int_{t_0}^t dt, \hat{H}'_I(t) |\Psi_I(t_0)\rangle$$

If this series converges quickly, you can truncate the series and predict the transition amplitude between any two states of the unperturbed Hamiltonian H_0 , $|i\rangle \rightarrow |f\rangle$.

This amplitude is written as:

$$A_{f \leftarrow i} = \langle f | \Psi_I(t) \rangle \quad \text{where } |\Psi_I(t_0)\rangle = |i\rangle$$

$$A_{f \leftarrow i} = A_{f \leftarrow i}^{(0)} + A_{f \leftarrow i}^{(1)} + A_{f \leftarrow i}^{(2)} + \dots \quad \text{where } A_{f \leftarrow i}^{(n)} = \langle f | \Psi_I^{(n)}(t) \rangle$$

We now see how to find $A_{f \leftarrow i}^{(1)}$ (one-photon amplitude).

$$A_{f \leftarrow i}^{(1)} = \langle f | \Psi_I^{(1)}(t) \rangle, \quad |\Psi_I^{(1)}(t)\rangle = \frac{1}{i\hbar} \int_{t_0}^t \hat{H}'_I(t_1) \underbrace{|\Psi_I(t_0)\rangle}_{=|i\rangle}$$

so, $A_{f \leftarrow i}^{(1)}$ is:

$$\begin{aligned} \langle f | \frac{1}{i\hbar} \int_{t_0}^t dt' \hat{H}'_I(t') | i \rangle &= \frac{1}{i\hbar} \int_{t_0}^t dt' \langle f | \hat{H}'_I(t') | i \rangle \\ &= \frac{1}{i\hbar} \int_{t_0}^t e^{i(E_f - E_i)t'/\hbar} \langle f | \hat{H}'_I(t') | i \rangle dt' \end{aligned}$$

where we define $\omega_{fi} = \frac{E_f - E_i}{\hbar}$ and $H'_{fi}(t) = \langle f | \hat{H}'_I(t') | i \rangle$

$$A_{f \leftarrow i}^{(1)} = \frac{1}{i\hbar} \int_{t_0}^t e^{i\omega_{fi}t'} H'_{fi}(t') dt'$$

We consider the case where the perturbation is 0 at times $t' < t_0$ and $t' > t$, e.g. a pulse of radiation that starts at t_0 and ends at t . Apparently this lets us integrate over $-\infty$ and then replace the integral with the Fourier transform of $H'_{fi}(t)$.

$$\text{In general, } \tilde{f}(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} f(t) dt \quad \text{for } f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \tilde{f}(\omega) d\omega$$

$$\text{so, } A_{f \leftarrow i}^{(1)} = \frac{1}{i\hbar} \tilde{H}'_{fi}(\omega_{fi}).$$

But what is $\hat{H}'(t)$ in the first place? We assume $\hat{H}'(t) = F(t) \hat{\mathcal{O}}$ where $F(t)$ is a time-dependent field factor and $\hat{\mathcal{O}}$ is time-independent.

$$\text{This means } H'_{fi}(t) = \langle f | F(t) \hat{\mathcal{O}} | i \rangle, \quad \tilde{H}'_{fi}(\omega_{fi}) = \langle f | \tilde{F}(\omega_{fi}) \hat{\mathcal{O}} | i \rangle$$

or, since at a fixed ω_{fi} , $\tilde{F}(\omega_{fi})$ is fixed, $\tilde{H}'_{fi}(\omega_{fi}) = \tilde{F}(\omega_{fi}) \langle f | \hat{O} | i \rangle$

$$A_{fi}^{(1)} = \frac{1}{i\hbar} \langle f | \hat{O} | i \rangle \tilde{F}(\omega_{fi}) = \frac{1}{i\hbar} \mathcal{O}_{fi} \tilde{F}(\omega_{fi})$$

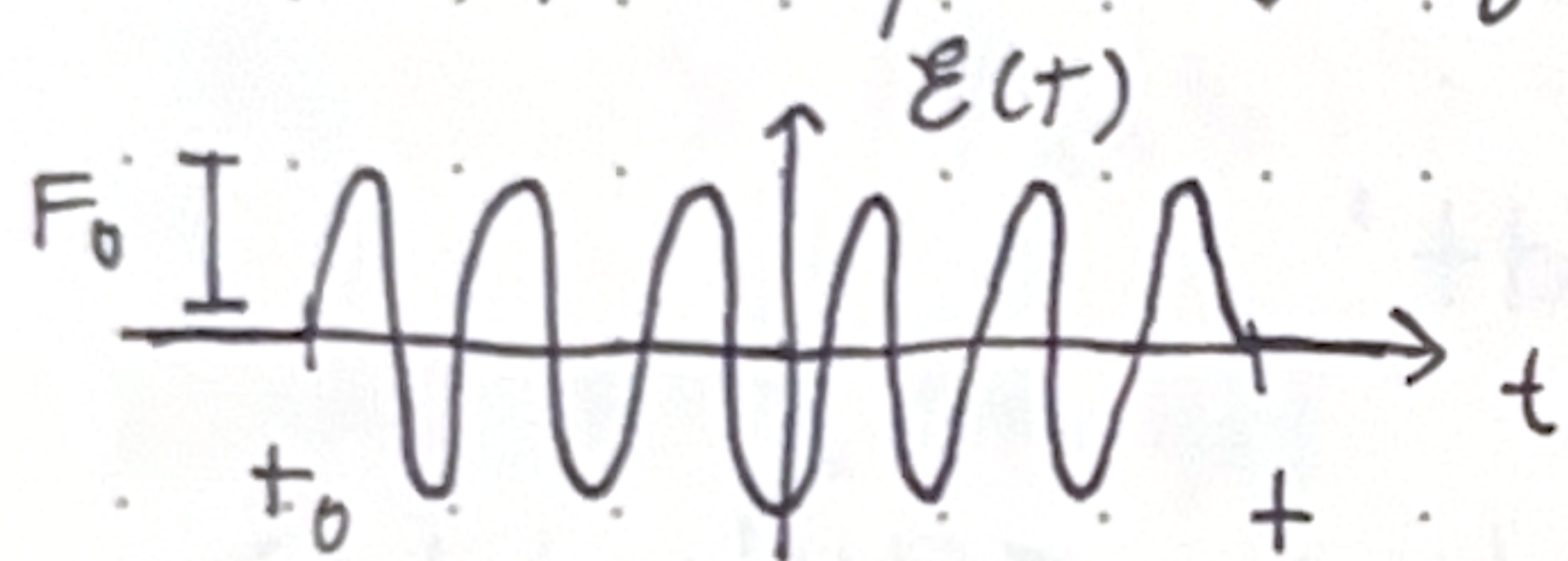
In length gauge specifically, non-interaction representation perturbation $\hat{H}'(t) = -\vec{E}(t) \cdot \hat{\vec{\mu}}_e$ where $\hat{\vec{e}}$ is the unit vector describing the polarization of the field, and:

$$-\vec{E}(t) = \text{our } F(t) \quad \text{and} \quad \hat{\vec{e}} \cdot \hat{\vec{\mu}}_e = \text{our } \hat{O}$$

For other gauges, this is different, but you can still write $\hat{H}'(t)$ as a product of $F(t)$ and $\hat{O}$.

Keep in mind that the point of finding $A_{fi}^{(1)}$ is to find $P_{fi}^{(1)}$ or the probability of this transition happening at all, to first order.

Specifically in the case where we have a pulse/perturbation of constant amplitude F_0 :



For duration $T = t_1 - t_0$ and clearly $F(t) = F_0 \cos(\omega_0 t + \phi)$

$$\begin{aligned} \text{We can write } F(t) &= \frac{1}{2} F_0 e^{i\phi} e^{i\omega_0 t} + \frac{1}{2} F_0 e^{-i\phi} e^{-i\omega_0 t} \\ &= F_{\omega_0} e^{i\omega_0 t} + F_{\omega_0}^* e^{-i\omega_0 t} \end{aligned}$$

As we said before, since this pulse is restricted to a finite time interval, we can apply the Fourier transform.

$$\begin{aligned} \tilde{F}(\omega) &= F_{\omega_0} \int_{-T/2}^{T/2} e^{i(\omega_0 + \omega)t} dt + F_{\omega_0}^* \int_{-T/2}^{T/2} e^{i(\omega - \omega_0)t} dt \\ &= 2F_{\omega_0} \frac{e^{i(\omega_0 + \omega)T/2} - e^{-i(\omega_0 + \omega)T/2}}{2i(\omega_0 + \omega)} + F_{\omega_0}^* \frac{e^{i(\omega - \omega_0)T/2} - e^{-i(\omega - \omega_0)T/2}}{i(\omega - \omega_0)} \\ &= 2F_{\omega_0} \frac{\sin(T/2(\omega_0 + \omega))}{\omega_0 + \omega} + 2F_{\omega_0}^* \frac{\sin(T/2(\omega - \omega_0))}{\omega - \omega_0} \end{aligned}$$

Where the amplitude is a function of T . Note that for large values of T , $\frac{\sin(x)}{x}$ for $x \propto T$ looks like a delta function, a demonstration I leave for digital notes.

Thus $\tilde{F}(\omega) = 2\pi F_{\omega_0} \delta_T(\omega_0 + \omega) + 2\pi F_{\omega_0}^* \delta_T(\omega - \omega_0)$

where:

$$\delta_T(\omega) = \frac{\sin(\omega T/2)}{\pi \omega} \quad \text{and} \quad \lim_{T \rightarrow 0} \int f(\omega) \delta_T(\omega) d\omega = f(0)$$

The full expression for $A_{f \leftarrow i}^{(1)}$ is:

$$A_{f \leftarrow i}^{(1)} = \frac{2\pi}{i\hbar} \langle f | \hat{V} | i \rangle \left[\underbrace{F_{\omega_0} \delta_T(\omega_{fi} - \omega_0)}_{\text{absorption}} + \underbrace{F_{\omega_0}^* \delta_T(\omega_{fi} + \omega_0)}_{\text{stimulated emission}} \right]$$

Notice that this implies the energy of the field must match the needed transition energy exactly for the transition to happen. This leads to the conclusion that matter can only absorb energy in quantized units.